Erdős Projects — Fall 2026 Executive Summary

Project: Predicting YouTube Shorts Virality

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GitHub: `https://github.com/Erdos-Projects/fall-2026-viral-videos`

Overview:

Short-form videos can become popular very quickly, yet it is difficult to know which uploads will achieve unusually high view counts. This project asks: *What makes a YouTube Short go viral, and can we predict it early?* We investigate how early viewing activity, engagement, creator characteristics, content metadata, and posting time relate to later popularity. Our goal is to identify promising videos six hours after publication, when an early prediction may be useful to creators, agencies, and platforms.

Objective:

We predict whether a video will be in the **top 10% of US and Canadian videos by views at approximately 72 hours**. The virality threshold is calculated from US/Canadian *training* videos, rather than from the test set. The main objectives are:

1. Identify early signals associated with high view counts at 72 hours.
2. Compare simple six-hour view-based rankings with machine-learning classifiers.
3. Evaluate predictions on newer videos from creators not seen in training.

Our primary performance metric is **average precision (PR-AUC)**, supported by ROC-AUC and precision and recall when flagging the top 10% of predictions.

Methodology:

Data Collection and Pre-Processing:

- Collected YouTube Data API observations at roughly hourly intervals between October 12 and November 1, 2026: **30,709 videos**, **15,687 channels**, and **2,168,782 engagement snapshots** across 440 files.
- Combined video metadata, channel information, channel-reported country, and time-stamped views, likes, and comments. Selected outcome observations nearest 72 hours (within ± 2 hours).
- Formed a six-hour modelling cohort of **25,345 videos**, including **2,781 from US/Canadian channels**; inspected missing values, snapshot coverage, and data revisions.

Feature Engineering:

- **Early performance:** log six-hour views and log view growth from three to six hours.
- **Engagement and creator:** likes, comments, missing-like indicator, subscribers, channel age, and historical channel statistics.
- **Content and timing:** duration, hashtag count, title length, emoji count, and cyclically encoded publication hour.
- Restricted video observations to the six-hour prediction cutoff; masked channel statistics first collected after that cutoff.

Machine Learning and Evaluation:

- Compared logistic regression, decision trees, random forests, and XGBoost across ten feature-set and training-population experiments.
- Used five channel-grouped cross-validation folds, repeated three times, for model development. All models were validated on US/Canadian videos, with some trained on worldwide data.
- Reserved the newest videos for a chronological, channel-disjoint test set, with a 72-hour exclusion interval preceding the test period.

Results and Advanced Methods:

Exploratory Findings:

- Six-hour views and 72-hour views had a Spearman rank correlation of approximately **0.83**, making early viewing activity a strong predictor.
- Metadata-only models performed substantially worse than models using early viewing and engagement signals.
- Feature ablations suggested that engagement adds useful information, but its incremental benefit is smaller once early views are included.

Model Comparison (Development Cross-Validation):

Model	Training data / features	Mean PR-AUC
Logistic regression	Worldwide / all	0.713
Decision tree	Worldwide / all	0.648
Random forest	Worldwide / all	0.732
XGBoost	**Worldwide / all**	**0.752**

The selected XGBoost configuration (maximum tree depth 3) achieved **0.752 ± 0.114 PR-AUC** and **0.937 ± 0.028 ROC-AUC** across development folds (mean ± standard deviation).

Final Held-Out Evaluation:

The final test contained **562 US/Canadian videos**, of which **48** met the training-defined virality threshold. The selected XGBoost classifier was compared with ranking videos by six-hour views alone.

Metric	Six-hour views baseline	XGBoost
Average precision (PR-AUC)	0.685	**0.757**
ROC-AUC	0.880	**0.911**
Precision at top 10%	0.561	**0.614**
Recall at top 10%	0.667	**0.729**
Viral videos identified	32 of 48	**35 of 48**

Conclusions and Future Directions:

Conclusions:

- **Early views provide a strong baseline.** A video's first six hours contain substantial information about its popularity at 72 hours.
- **Additional features offer a modest improvement.** XGBoost improved held-out PR-AUC from 0.685 to 0.757 and identified three more viral videos among the top 10% flagged.
- **Worldwide training helped development performance,** even though the evaluation focused on US/Canadian videos. These are predictive associations, not evidence that the features cause virality.

Limitations and Future Directions:

- The dataset contains search-discovered Shorts *candidates*, not independently verified Shorts, and covers only a short time period.
- Some engagement counts were missing or revised; 72-hour outcomes were estimated using nearby snapshots. A 72-hour publication-time embargo alone does not guarantee every training outcome was observed before testing began.
- Longer collection periods, stricter temporal validation, and additional prediction horizons (e.g., three hours) could strengthen generalization evidence.